

# NGO Data Engineering Pipeline

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# Executive Summary & Project Objective

1

## Goal

Transform raw NGO operational datasets — Beneficiary Cases & Service Logs — into an audit-ready, centralized analytical data platform.

2

## Architecture

Medallion Architecture (Bronze → Silver → Gold) built on Databricks, PySpark, and Delta Lake for scalable, layered data processing.

3

## Impact

Automated quality assertion logging, operational dashboard reporting, and full workflow idempotency across all pipeline runs.

# End-to-End Medallion Architecture

The Medallion Architecture structures data refinement into three progressive layers — each adding quality, governance, and analytical value on top of the previous.



## Bronze Layer — Ingestion

Raw CSV ingestion without schema modification, enriched with metadata columns `_ingested_at` and `_source_file` for full lineage tracking.



## Silver Layer — Cleansing & Quality

Strict null filtering on Primary Keys (`case_id`, `service_id`), date type casting, and PySpark Window-based deduplication for clean, trustworthy records.



## Gold Layer — Analytics & KPIs

Aggregated business datasets including daily service trends (`gold_daily_services`) and a 360-degree beneficiary view (`gold_case_360`).

# Databricks Workflow Orchestration

The pipeline is fully orchestrated as a single Databricks Workflow — **NGO\_Pipeline\_Orchestration** — ensuring sequential, dependency-aware execution with zero manual intervention and idempotent re-run safety.



4

## Notebook Stages

Sequential execution: 00\_config →  
01\_bronze → 02\_silver →  
03\_gold



2m7s

## Completion Time

Serverless execution completed  
successfully in 2 minutes 7 seconds  
with zero errors.



0

## Pipeline Errors

Clean run every time thanks to Delta  
Lake overwrite modes and  
deduplication logic ensuring full  
idempotency.



∞

## Idempotent Re-Runs

Safe to re-trigger at any time — Delta  
Lake overwrite semantics and  
Window-based deduplication prevent  
data duplication.

# Automated Data Quality Framework

Quality is enforced programmatically in the Silver layer and logged automatically into `gold_quality_summary` — creating a self-documenting audit trail with every pipeline run.

## Null Case IDs Check

Rule: `null_case_ids` in `silver_cases`. Identifies and logs any records missing a primary `case_id` before they can propagate to the Gold layer.

## Duplicate Cases Check

Rule: `duplicate_cases` in `silver_cases`. Enforces uniqueness of beneficiary case records using PySpark Window partitioning over ingestion timestamps.

## Null & Duplicate Service IDs

Rules: `null_service_ids` and `duplicate_services` in `silver_services`. Mirrors the same rigorous checks applied to the service log dataset for full coverage.

# Technical Deep-Dive & Architecture Decisions

Three foundational engineering decisions ensure this pipeline is production-grade, scalable, and fully governable — from deduplication logic to streaming readiness and operational lineage.

## Idempotency & Deduplication

Handled via PySpark Window partitioning over ingestion timestamps:

```
Window.partitionBy("primary_key").orderBy(col("_ingested_at").desc())
```

 — safe to re-run without duplicating records.

## Governance & Auditing

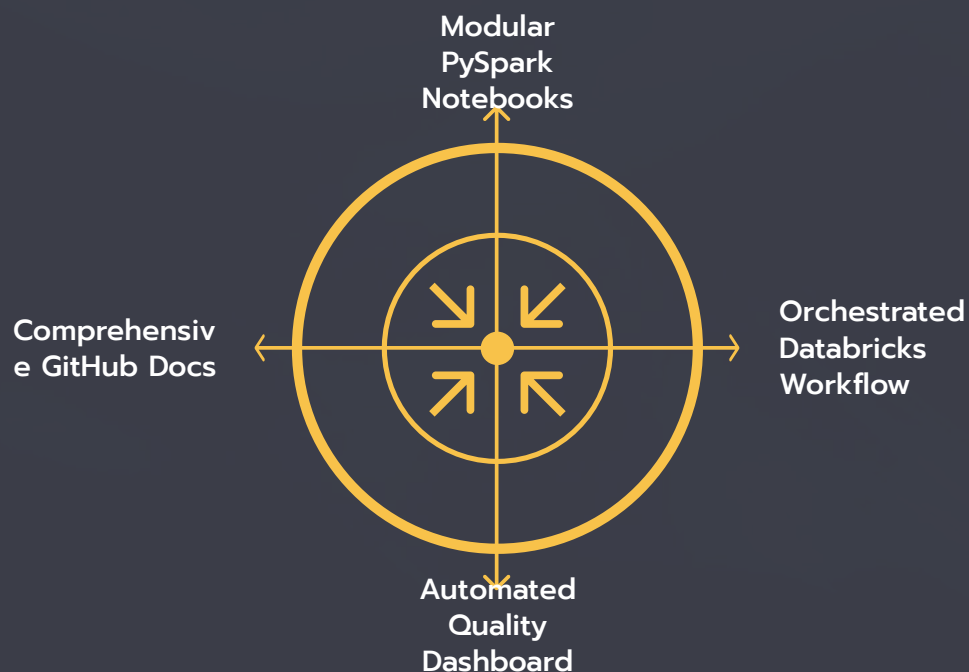
Filtered rows are tracked via quality failure logs to preserve complete operational lineage — every rejected record is accounted for in `gold_quality_summary`.

## Streaming Scalability

Architecture is ready to scale from batch to continuous ingestion via Databricks Auto Loader (`spark.readStream`) and Delta `MERGE INTO` upserts — no redesign required.

# Business Outcomes & Next Steps

The pipeline delivers four modular, production-ready artefacts alongside a clear roadmap toward fully observable, alert-enabled operations integrated with the broader Databricks ecosystem.



Each deliverable is independently versioned and documented, enabling rapid onboarding and future extensibility by any engineering team member.

1

## Slack / Email Webhooks

Integrate pipeline failure alerts via Slack or Email webhooks to enable instant incident response and reduce mean time to recovery.

2

## Databricks System Tables

Advanced observability using Databricks System Tables for query history, cost tracking, and pipeline health monitoring across all environments.

3

## Continuous Ingestion

Evolve from scheduled batch runs to real-time streaming ingestion using Auto Loader and Delta MERGE INTO as NGO data volumes grow.

# Pipeline Monitoring & Quality KPIs

Every pipeline run produces measurable quality signals. Tracking these KPIs across runs enables proactive governance, continuous improvement, and confident data delivery to NGO stakeholders.

## Null Record Rate

Track the percentage of records rejected for missing Primary Keys (`case_id`, `service_id`) across each pipeline run to identify upstream data quality issues.

## Gold Layer Row Count

Validate expected record volumes in `gold_daily_services` and `gold_case_360` on each run to catch silent data loss or schema drift early.

## Deduplication Volume

Monitor the count of duplicate rows removed in the Silver layer over time. A sudden spike indicates a source system issue requiring upstream investigation.

## Quality Summary Dashboard

The automated `gold_quality_summary` table feeds real-time dashboards — giving data engineers and NGO managers a live view of pipeline health and data trustworthiness.

